

MECHANICAL SPECIFICATION
FOR OFFICE REFURBISHMENTS
AT
STANWELL POWER STATION
FOR
STANWELL CORPORATION

ARCHITECTS

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TABLE OF CONTENTS

1.0	EXTENT OF WORK	3
2.0	DRAWINGS	4
3.0	WORK NOT INCLUDED IN MECHANICAL SERVICES	4
4.0	DESIGN CONDITIONS.....	4
5.0	PERFORMANCE GUARANTEE.....	5
6.0	NOISE LEVELS.....	5
7.0	STANDARDS	5
8.0	SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS	5
9.0	SPLIT AIR CONDITIONING UNITS	5
10.0	PLINTHS AND EQUIPMENT BASES.....	6
11.0	AIR FILTERS	7
12.0	FILTER/MIXING BOXES	7
13.0	ELECTRIC HEATERS	8
14.0	DUCTWORK.....	8
15.0	PIPEWORK AND FITTINGS.....	10
16.0	VARIABLE AIR VOLUME DAMPERS	12
17.0	FLEXIBLE DUCTWORK	12
18.0	FLEXIBLE CONNECTIONS.....	13
19.0	CUSHION HEADS	13
20.0	DAMPERS.....	13
21.0	AIR DIFFUSION EQUIPMENT	14
22.0	INSULATION	14
23.0	VENTILATION SYSTEMS	15
24.0	CONTROLS AND CONTROL REQUIREMENTS	15
25.0	ELECTRICAL	17
26.0	PAINTING AND CORROSION PROTECTION	21
27.0	TESTING AND COMMISSIONING	22
28.0	DEFECTS LIABILITY PERIOD	23
29.0	SERVICE AND MAINTENANCE	23
30.0	SHOP DRAWINGS.....	23
31.0	OPERATION AND MAINTENANCE MANUALS	24
32.0	SCHEDULE OF EQUIPMENT MINIMUM CAPACITIES	25

1.0 EXTENT OF WORK

The extent of work generally comprises the design, supply, delivery, installation, testing, commissioning, servicing and maintaining for a period of twelve months of:

- five (5) new vertical discharge split air conditioning units, two (2) units to be complete with VAV zone control, and three (3) units to be complete with CO2 control;
- building management system (BMS) for the new building air conditioning units;
- one (1) new toilet exhaust system;
- associated ductwork, filters, grilles, louvres, and associated items;
- units to be complete with all controls;
- all electrical wiring and controls associated with the above;
- all required refrigerant and condensate pipework;
- painting of equipment as hereinafter specified;
- all lifting and hoisting of equipment;
- plinths for mounting of equipment;
- for existing buildings, internal cleaning of all existing air conditioning ductwork;
- for existing buildings, replacement of all existing air conditioning registers, filters and flexible ductwork;
- for existing buildings, internal cleaning of all existing toilet exhaust ductwork and grilles;
- workshop drawings;
- as built drawings;
- operating instructions and maintenance manuals;
- service and maintenance for the Defects Liability Period;
- all other work, whether especially mentioned or not, necessary to provide a complete operative installation and for the effective and efficient running of the plant and within the boundaries of the work of this contract.

2.0 DRAWINGS

The following drawings form part of this specification and are intended to show the general arrangement and the extent of work to be carried out.

Drawing No. Title

M-01	Part Site Plan, Legend & Notes
M-02	Registry Building – Air Conditioning and Mechanical Ventilation
M-04	Workshop Building – Air Conditioning and Mechanical Ventilation
M-04	Warehouse Building – Air Conditioning and Mechanical Ventilation
M-05	New Building – Air Conditioning and Mechanical Ventilation

Any information involving accurate measurement of the building shall be taken from the architectural and structural drawings and on site.

The set-out and final location of all items associated with the works shall be determined on site in co-ordination with other services and architectural details prior to installation.

3.0 WORK NOT INCLUDED IN MECHANICAL SERVICES

The following works and/or provisions associated with Mechanical Services shall be provided by other trades at no cost to this contractor.

It shall be this Contractor's responsibility to advise and ensure that these works and/or provisions are carried out to their satisfaction and to the satisfaction of any other properly constituted authority having jurisdiction over the project.

- positioning and fixing of all sleeves, equipment and holding down bolts etc. (This Contractor shall supply all sleeves and provide drawings giving the exact details, size and position of these items so that they may be accurately located).
- installation of door grilles and external grilles to plantroom (grilles to be supplied by this Contractor);
- the provision of all access panels required in the building structure including the necessary details of the openings, access to valves, etc;
- the provision of suitable condensate drain points as indicated on the drawings;
- the provision of electrical supplies as elsewhere detailed herein.

This Contractor shall confirm that the above are included in the building works.

4.0 DESIGN CONDITIONS

The cooling and heating loads of the building shall be based on the following design conditions.

Summer

Outside:	37.1°C DB	26.3°C WB
Inside:	23°C DB	50% R.H. (design figure only)

Winter

Outside:	9.8°C DB
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Inside: 21°C DB

5.0 PERFORMANCE GUARANTEE

This Contractor by entering into this contract shall be deemed to guarantee the equipment selected and the installation as being capable of the capacity and performance as required by this specification and tender drawings.

All capacities, sizes, equipment, controls and the like are minimum provisions only and this Contractor as part of the work of this contract, shall supply all extra materials, equipment, etc. and do all extra work as necessary to meet the design conditions as elsewhere specified.

6.0 NOISE LEVELS

Sound pressure levels due to components of the mechanical ventilation installation shall be in accordance with AS 2107 and AS 1055 and shall not exceed NR35 in occupied areas.

Tenderers shall consider the above noise level requirements when selecting their equipment, to ensure that the stated levels can be maintained. Should a tenderer consider that additional acoustic treatment or noise reduction equipment or methods be used to attain the specified levels then he should include for this in his tender.

7.0 STANDARDS

All works under this contract shall be performed in accordance with the latest edition of the relevant Australian Standards, the Building Code of Australia and all other Authorities having jurisdiction over the work.

8.0 SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS

Arrange all components, other than service items exempt in AS 1170.4, to resist seismic loads determined in accordance with AS 1170.4. Calculations shall be completed by a Structural Engineer and shall be specific to the element being considered. Securely fix all plant and equipment to the building structure. Do not rely on gravity and/or friction to resist seismic forces.

The spacing of lateral and vertical braces shall be determined for each service however, a maximum spacing of 2m between supports shall be provided.

Locations of all braces shall be coordinated with all services to avoid clashes.

Seismic restraints calculated in accordance with AS1170.4 are the minimum requirement and shall be provided in addition to specific vibration isolation and supports nominated elsewhere herein.

9.0 SPLIT AIR CONDITIONING UNITS

The air conditioning units shall be cooling only split air conditioning units for AC-2 & AC-4; and reverse cycle split air conditioning units for AC-1, AC-3, & AC-5. They shall be fully factory assembled, with sectionalised construction fabricated from heavy gauge galvanised sheet metal, rigidly braced and bolted together. Panels shall be easily removable to allow access for servicing and maintenance purposes.

The air conditioning units shall comply with the relevant sections of the Australian Government Minimum Energy Performance Standards (MEPS), and the National Construction Code (NCC).

Casings and all accessories, with the exception of the coil, shall be finished in protective weatherproof enamel finish.

All cooling sections of units shall be fully internally insulated with minimum 12mm thickness of closed-cell elastomeric foam insulation faced with aluminium foil.

The units and associated fan motors shall be selected for quiet operation so as to comply with the design acceptable noise levels.

Fans shall be statically balanced and selected to obtain maximum performance for the duty involved.

The refrigeration system shall comprise hermetic compressors operating on R410A. A minimum of two stages of capacity control with two refrigeration circuits or at least one digital scroll compressor shall be provided for units greater than 25 kW_r.

Infinite capacity control shall be provided for all refrigeration systems via inverter drive speed control or digital scroll compressors.

Cooling and condenser coils shall have copper tubes with aluminium fins.

Cooling coils shall be provided with condensate tray constructed from stainless steel or marine grade aluminium with access panels upstream and downstream of the coils to facilitate cleaning.

Condenser fans shall be of the low noise type to comply with adjacent maximum noise level requirements.

Protect condenser coils against hail damage.

The air conditioning units shall be fully internally factory wired.

Run an insulated PVC condensate drain from the unit to discharge over a tundish with 25mm air gap as shown on the accompanying drawing. Provide a trap in the condensate line adjacent to the units. All condensate pipework shall be insulated with minimum 13mm Armaflex or approved equal over the full length of the line to the discharge point.

Floor mounted air handling units shall be mounted on cup-located open spring supports with rubber isolated bolt holes, similar to Embelton XL series.

Condensing units and rooftop package air conditioning units shall be mounted on a double layer of neoprene waffle pads.

The units shall be selected to meet the minimum capacity given in the Schedule of Air Conditioning Unit Minimum Capacity with condenser air at 35°C DB.

10.0 PLINTHS AND EQUIPMENT BASES

The Mechanical Contractor shall fabricate a 100mm high plinth surround, fix it into position and provide concrete filling.

The surround shall consist of minimum 1.6mm fabricated steel with welded joints with all edges turned over to ensure a neat surround. The surround shall be treated for corrosion protection and finally painted as elsewhere specified.

Plinths to be provided for all indoor and outdoor units.

11.0 AIR FILTERS

(a) Panel Filters

The conditioner unit shall be fitted with panel filters containing 100mm thick dry disposable media.

The filters shall have an F5 rating and conform to the following:

- minimum efficiency to AS 1324.2 No. 1 test dust. 20%;
- average efficiency to AS 1324.2 No. 1 test dust. 40-60%.
- average arrestance to AS 1324.2 No. 4 test dust. 95%.

Dust holding capacity shall not be less than 580g/m² when tested in accordance with AS 1324.2 at a final resistance of 250pa.

Filter shall be similar to Email Interfold TP 100.

Filter banks shall be adequately sized and arranged such that the face velocity does not exceed 2.5m/s with the system operating at maximum air quantity.

(b) General

Provide suitable hinged doors with heavy duty die-cast catches for access for cleaning and removal of filters.

Filters shall be replaced at not more than yearly intervals or as recommended by the filter manufacturer. This Contractor shall install a new set of filters media before the end of the Defects Liability Period.

12.0 FILTER/MIXING BOXES

The filter/mixing box shall be constructed in 1.2mm thick galvabond, suitably braced and stiffened.

Internally insulate throughout as per INSULATION section and face with perforated sisalation. Provide perforated galvanised metal facing over all surfaces of the insulation where personnel access is required.

The air filter removal section shall be faced with 1.0 mm thick perforated sheet metal for mechanical protection.

Folded sheet metal access doors shall be provided where shown on the drawings and insulated as for the remainder of the filter mixing box and faced with perforated sheet metal.

Doors shall be complete with 2-off 'EFCO' adjustable hinges and 2-off 'LARK' L1068 die-cast handles located 300mm from the top and bottom of the door. Handles shall be of the double-acting type to allow access from either side of the door.

Doors shall have gaskets comprising a continuous 6mm diameter rod tack welded all around the perimeter to substantial door casing and this rod shall close onto a continuous neoprene rubber gasket bonded into a 1.6mm thick sheet metal housing fixed to the filter/mixing box casing.

13.0 ELECTRIC HEATERS

The heating elements shall be of the totally enclosed finned tubular type, the surface of which shall not exceed 400°C when tested in accordance with AS 3102. All heater banks must fully comply with AS3102.

The heater bank shall be located properly supported in an accessible position for easy service and removal and duct sized to give an air velocity over the heaters of not more than 3.5 m/s. Heaters shall be so arranged to provide an even temperature distribution across the air stream.

The heater bank shall be protected by circuit breakers and shall be electrically interlocked with the respective fan motor so that the elements cannot be energised unless the fan motor is in operation.

A manually reset safety thermostat shall be installed additionally to the abovementioned electric interlock as a precaution against overheating in the duct. The thermostat will de-energise the heater when the air in the duct downstream within 250mm of the heater elements reaches a temperature of 120°C.

An air supply switch shall be installed in the upstream side of the conditioner to interrupt electric power supply to the heaters should power supply to the conditioners cease.

Where the air supply failure switch is incorporated within the air handler unit casing the direction of airflow shall be clearly and permanently marked on the casing.

The use of vane switches is not permitted.

The elements shall be so arranged that wiring connections to them are contained within a sheet metal box external to the duct section housing the heater.

Each bank of heaters 3 kW and over shall be balanced over all three phases.

Each heater bank over 6kW shall have a minimum of two stages of heating or SSR proportional control.

The casing of the heater shall be internally insulated for a distance of not less than 300mm upstream and downstream of the heater with a non-combustible material of thermal conductance not greater than 30 W/sq.mK at 100°C.

14.0 DUCTWORK

General

The specification and drawings indicate the extent and general arrangement of the ducting system. If any departures from the contract drawings are deemed necessary by this Contractor, details of such departures and reasons therefore shall be submitted for approval.

Approved departures shall be made at no extra cost to the Proprietor unless departures involve more work and are due to changes in design or requirements.

Where ductwork is exposed, care must be taken to ensure equal panel lengths and a neat and workmanlike appearance.

All sheet metal ductwork shall be constructed, tested and commissioned to comply with; AS 1668.1:2015, AS 4254.2:2012 and the Sheet Metal and Air Conditioning Contractor's National Association Inc. (SMACNA) Manuals 'Duct Construction Standards' except that snap lock duct seams are not acceptable.

Ductwork shall be constructed to suit a static pressure of 500 Pa.

Design resistance to air flow shall not exceed 1pa/m.

Access panels are required adjacent to all internal items in ductwork requiring inspection, cleaning and service. Return air, exhaust air and outside air ductwork shall have access panels at not more than four (4) metre centres (limit reach to 2 metres). Supply air ductwork shall have an access panel in the plant area plus an access panel for each zone in the ceiling space and easily removable stop end in the end of each branch duct.

Kitchen exhaust ductwork shall have access panels for cleaning and service. They shall be located at each change in direction, at not more than three (3) metre intervals in horizontal runs (limit reach to 1.5 metres) and at the base of vertical risers.

An access panel shall also be provided directly below each roof penetration.

Where ductwork penetrates the exterior of the building, security screens shall be installed in the ducts to prevent access through the ductwork.

Ductwork shall be constructed from galvanised metal sheet or approved equal.

All dimensions are clear internal sizes which in the case of internally lined ducts are inside the insulation.

Ductwork shall be free of any waves or buckles and all sheets, except in the case of round ductwork, shall be machine bent to ensure neat, square and accurate fabrication

Unless otherwise specified or shown on the drawings, all changes in duct size shall be made in a gradual manner with a total angle of divergence not exceeding 15 deg. Wherever possible, bends shall be made with a radius not less than duct width in the plan of the bend, but where this is not possible due to space limitations small radius bends may be used with properly designed and approved turning vanes.

All ductwork shall be adequately supported adjacent to all flexible connections.

All ductwork shall be securely supported so as to prevent vibration or movement, but arranged to allow expansion due to thermal stresses without distortion of the ductwork, rupture of insulation or damage to supporting structure.

Ensure that all dampers are free from vibration and rattles under all operating conditions.

Round or oval spigots shall be used for flexible ductwork. All spigots shall be fitted with butterfly dampers.

15.0 PIPEWORK AND FITTINGS

All pipework, whether shown on the drawing and/or specified, shall be installed in an approved manner to meet structural, architectural and functional conditions and shall not interfere with the work of other trades.

All pipes sizes shall be as per manufacturer's recommendations for the duty required and shall be subject to approval.

All pipework shall be constructed, installed and supported so that it is free from excessive stressing due to weight from its contents plus its own dead weight, dynamic forces due to liquid movement and expansion and contraction due to change in temperature.

Care shall be taken to ensure neat and workmanlike appearance as well as true alignment and grade.

All pipes and fittings shall be thoroughly cleaned before erection and all scale, burrs, sand, slag and other obstructions shall be removed.

All horizontal pipework shall be installed with a constant grade of not less than 25 mm per 30 metres either towards a drain point or towards a vertical high point.

In designing the pipework, care must be taken to ensure that sufficient unions or flanged joints are installed to allow satisfactory removal of piping, valves and fittings for inspection or repair.

All open ends of pipework during erection shall be properly sealed by a metal or plastic cap at the end of each day's work or at the end of each section of work.

Where pipings pass through walls or the roof, the openings shall be flashed weatherproof in an approved workmanlike manner.

Should any damage be caused to building structure or finishes by leaks from piping systems, the cost of repair shall be borne by this Contractor.

Pipe Materials

The following piping materials shall be considered the minimum acceptable standard.

- Drains
uPVC.
- Refrigerant Piping
Hard drawn copper to AS.1571:1995.

Pipe Supports

Refrigerant pipework shall be supported via a 'Unistrut' or equal channel type pipe support system. Channel size shall be as per manufacturer's recommendations. Pipework shall be secured to the channel via proprietary pipe clamps. Provide rigid pipe insulation inserts under all clamps to prevent compression of the insulation.

Supply all hangers, springs, supports, beam clamps, inserts and mounting devices, including ancillary steelwork for connection to structure, for the satisfactory support of all piping installed under this contract.

Supports for vertical pipe lines shall be designed so that the weight of the pipe is positively supported in a way that does not depend on a friction grip onto the pipe.

Maximum spacing between horizontal supports shall be as follows:

Copper

Diameter:	40 & below	50 & over
Span, m:	1.8	3.0

Hanger Rods

Diameter of pipe:	20 & below	25-50	65 & over
Diameter of rod mm:	6.5	10	12

A hanger shall be placed not more than 600mm from each change of direction, preferably in the side with the longest run.

All pipe supports and clips shall be of approved design.

Pipe hangers or brackets shall be properly isolated where necessary to prevent noise and vibration transmission and special care shall be taken in making piping connections to pumps and all other rotating equipment.

In particular, piping connected to spring mounted equipment such as compressors and cooling coils etc. shall be laid out with bends as necessary so as to have sufficient flexibility to allow for movement of the equipment and expansion or contraction of piping and shall be carried on supports incorporating springs for a distance of not less than 10 m. Because of expected movement of the condensing set bases, refrigerant pipework must bend to allow movement in all three directions.

The springs shall have deflection under operating conditions of at least 25 mm. Spring diameters shall be no less than 0.8 times the compressed height of the spring at design load. Springs shall have a minimum additional travel to solid equal to 50% of the design deflection. Each spring shall have a pointer and scale for indicating the spring deflection.

Bends and Tees

All bends shall be pulled in an approved manner and no appreciable flattening of the pipe discernible. Bends shall be of the long radius type.

Tees in copper lines shall be capillary type using silver solder.

Testing

Pneumatic pressure tests using dry nitrogen shall be carried out on all pipework at a minimum pressure of 2600 kPa (gauge) except suction pipework which shall be carried out at a minimum pressure of 1400 kPa (gauge). R410a systems shall be tested to 3700kPa or as per manufacturer's recommendation. The tests shall be held for twenty four (24) hours.

If there is any pressure drop during the test period, the pipework shall be checked for leaks with soap suds and after repairs have been made the test shall be repeated.

After pressure testing, a vacuum test equivalent to a minimum of 300 microns shall be maintained for a period of twenty four (24) hours. If there is any rise during the vacuum tests, the pipework shall be checked for leaks and after repairs have been made the tests shall be repeated.

All joints in all pipework shall be exposed at the time of testing and pipework shall not be lagged prior to testing.

Refrigerant Pipe Insulation

Suction and liquid refrigerant lines shall be insulated with preformed elastomeric sponge equal to FRV ARMAFLEX or approved equal installed strictly as recommended by the manufacturer. Insulation shall be weatherproofed to approval.

Minimum refrigerant pipework insulation thickness shall be as follows:

less than 40mm diameter:	25mm
greater than 40mm diameter:	38mm

Provide removable top-hat section covers over all exposed refrigerant pipework and adjacent electrical conduits. Covers to be minimum 'Colorbond' to match adjacent finishes or aluminium checkerplate where foot traffic occurs. Care must be taken adjacent unit connections to ensure all pipework and insulation is completely shielded from sunlight.

Pipemarkers

All piping, whether exposed or concealed shall have permanent pipe markers ('Brady' or 'Safetyman' or equal) to identify piping and to indicate the direction of flow. Markers and the intervals at which they are applied shall be in accordance with AS.1345.

16.0 VARIABLE AIR VOLUME DAMPERS

Variable air volume dampers shall be of the pressure-dependent type, and consisting of contra-rotating dampers within the ductwork activated by a drive motor, controlled by a space temperature sensor.

17.0 FLEXIBLE DUCTWORK

Flexible ductwork shall be constructed and installed in accordance with AS 4254.1:2012. It shall be factory fabricated from a resilient helix of zinc coated spring steel wire and shall be internally lined with a tough aluminium laminate.

All air conditioning flexible ductwork shall be insulated with 40mm thick insulation complying with the INSULATION section of this specification, with a total minimum material R value of R1.0.

The maximum length of flexible duct in any run shall not exceed 6m.

Flexible ductwork shall be of the 4 zero type and be secured with approved taping and cable tie clamp over internal sleeve. The external sleeve shall be taped to the duct or cushion head to maintain the insulation thickness and vapour barrier with heavy duty 450 sisalation tape.

All flexible ductwork shall be supported from the roof or slab over. Provide suitable sling to support the flexible ductwork.

18.0 FLEXIBLE CONNECTIONS

For general purpose conditions all flexible connections shall be formed from "Coustibell" type "M" flexible, loaded, ignition retardant green vinyl or similar.

Flexible connections shall be provided on inlets and outlets of all fans and conditioner units where shown on the drawings.

Special care shall be taken to ensure air tight seals.

Connections shall have 100mm to 150mm length free of stress between the joined pieces of equipment arranged so as to permit renewal of fabric without disturbing ductwork or other plant.

Flexible connections on air conditioning unit supply air outlets shall be externally insulated with 50mm insulation. Insulation shall be taped to the duct / fan coil unit to maintain the vapour barrier with heavy duty 450 sisalation tape.

Fit sheet metal protective covers where exposed to weather.

19.0 CUSHION HEADS

All ceiling diffusers shall be fitted with internally insulated cushion head boxes. Ensure insulation extends to cover the full top section of the air register. All insulation shall comply with the INSULATION section of this specification. The insulation shall be covered with matt black perforated sisalation.

Boxes shall be secured to roof structure over, as applicable.

Lightweight cushion head boxes shall be constructed of 'P3' type sandwich panel comprising minimum 20mm thickness polyurethane foam board with embossed aluminium foil facing bonded to both sides. The panel shall achieve an overall insulation value of R1.5 and shall be tested to achieve a fire rating of zero in all four (4) regulatory indices when tested in accordance with AS/NZS1530.3:1999. The boxes shall be rigidly constructed with angle edge protection and all joints sealed and airtight.

20.0 DAMPERS

Dampers shall be constructed from galvanised steel or aluminium.

Galvanised steel blades shall be 1.6 mm minimum thickness and frames shall be 2.6 mm minimum thickness.

Aluminium blades and frames shall be 3.0 mm minimum thickness.

Shafts shall be Grade 304 stainless steel.

Bearings shall be porous bronze for outside air locations and nylon for internal locations.

Dampers shall be installed with blades horizontal and arranged for easy dismounting for maintenance purposes.

Damper blades shall be linked for ganged operation and free of slackness.

Dampers installed in ductwork shall have shafts extending through the ductwork with a locking quadrant.

21.0 AIR DIFFUSION EQUIPMENT

All equipment shall be of aluminium construction of the sizes as nominated on the drawings.

All ceiling diffusers shall be of the four way louvre faced design. All faces shall be of standard size or 600 x 600 for laying in a tee bar application. All supply air outlets shall have volume control dampers at the duct take off before the flexible ducting.

Thermally powered VAV diffusers shall be a complete VAV terminal and thermostat self-contained in a nominal 600 x 600 diffuser. They shall be thermally powered with one room thermostat/actuator and one changeover thermostat/actuator. External wiring or pneumatics shall not be allowed.

Ceiling mounted return air and exhaust grilles shall be of the egg-crate design. All return air and exhaust grilles shall be complete with opposed blade balancing dampers.

Wall mounted return air grilles shall be of the half chevron type with adjustable frames.

All door grilles shall be of the full chevron type with adjustable frames.

All equipment shall be finished to selected colours in accordance with the Client's colour scheme.

All sizes, especially those for tee bar application shall be verified with the Builder before ordering.

All equipment fitted with volume control dampers shall be capable of having the volume controls adjustable through the face of the grille, intake, register, etc.

The faces of all air diffusion equipment shall be easily removable for cleaning.

22.0 INSULATION

General

All insulation shall comply with AS 1668.1:2015 and the Sheet Metal and Air Conditioning Contractor's National Association Inc (SMACNA).

Air conditioning supply and return air ductwork shall be insulated as indicated on the drawing and minimum as follows:

- | | |
|--|-----------------------------------|
| - Ductwork within a roof space: | 75mm insulation (R2.0); |
| - External ductwork: | 100mm internal insulation (R3.0); |
| - Ductwork in a plantroom (including filter plenum): | 75mm insulation (R2.0); |
| - All other areas: | 50mm insulation. |

All insulation work shall be carried out by experienced tradesmen.

Where equipment is insulated, provision shall be made for a neat insulation finish around damper operating quadrants, splitters, adjusting clamps, access doors, valves and any similar operating devices.

All materials shall have a spread of flame index of zero and a smoke developed index not greater than 3 when tested to AS.1503 Part 3.

Ductwork Insulation

Internal insulation shall be rigid glasswool duct liner having a mass of 32 kg/cu.m and thermal conductivity of 0.031W/mK at 20°C, bonded to perforated 450 sisalation. In areas where personnel access is required perforated galvanised metal sheet shall be installed over the insulation. Fixing shall be by studs and clips on 300 mm spacings. Provide protective caps to fastening pins in accessible area of ductwork. All due care shall be taken to ensure that insulation circumferential joints are carried out in a workmanlike manner to prevent any cold tracking problems.

External insulation shall be glasswool duct wrap having a mass of 22 kg/cu.m and a thermal conductivity of 0.034 W/mK at 20 deg.C, faced with 450 sisalation. Fixing shall be by studs and clips on 300 mm spacings on bottoms of all ducts over 600 mm wide to prevent sagging. All other surfaces shall be strapped with 13 mm wide nylon straps and joints taped to suit with Bear Norton tape 250 grade having a width of 75 mm.

23.0 VENTILATION SYSTEMS

Fans shall be supplied to match the specified duties and space requirements as detailed on the drawings and indicated on the Fan Schedule.

Selection shall be made with due regard to operating efficiencies and noise ratings.

All exhaust systems shall be installed and operate in complete accordance with all relevant Codes and Authorities.

Toilet Exhaust Systems

The system shall be complete with all ductwork, intake grilles, fans, electrical work and controls etc. as generally shown on the drawing.

The systems shall comprise roof mounted axial or centrifugal fans with discharge cowl and complete with all accessories.

24.0 CONTROLS AND CONTROL REQUIREMENTS

General

All controls for the automatic control of the Mechanical Services shall be of the electronic type.

The control system shall be of an approved make and shall be installed (and maintained during the contract maintenance period) by an approved specialist organisation having all necessary personnel currently based in the local area.

All sensors, controllers and operators etc. shall be types suitable for maintaining temperatures within the limits as elsewhere specified.

The whole of the installation shall be entirely automatic in operation.

The whole of the installation shall automatically restart on restoration of power after a power failure.

Office Air Conditioning Systems (AC-2 & AC-4)

Each air conditioning unit shall be complete with an automatic start-up and shutdown, 365-day time clock with holiday programming. Provide a building management system with a communication output and online access for monitoring and adjustment.

Air conditioning units to be provided with manufacturer's optional high level interface systems.

All controllers shall be grouped together in a switchboard panel within mech store 1.

All space thermostats shall be suitable for wall mounting. Temperature adjustment shall be from the BMS system only. A suitable dead band shall be incorporated before heating elements are energised and VAV's operate.

For each zone the variable dampers shall be under the control of a modulating damper motor responsive to a zone space sensor positioned in the occupied space. All sensors shall be tamper proof.

Variable volume dampers shall be set up so that on reaching a minimum step air flow position, which shall not be less than 80% of the supply air, a dead band shall occur, before bringing on the heater bank as required.

A highest signal space sensor shall control the action of the compressor/s, so that the zone requiring the most cooling shall control the temperature of the air leaving the cooling coil. The remaining zones shall have their space temperatures maintained by the operation of the variable volume dampers either by reduced air quantities and/or reheat as required.

Provide space on the switchboard, to allow addition of controls for four additional zones for each switchboard.

Provide a remote control panel where shown on the drawings to allow for out of hours operation. It shall be a switch, labelled 'off/auto'. In the 'auto' position the system shall operate via the time clock control. Where it is not shown, provide near main staff entry point.

For after hours operation of each unit, a labelled push button shall energise an adjustable 0 - 120 minute run timer to control the length of time that the plant operates in the after hours condition. Remote control panel shall be complete with green "ON" light and red "FAULT" light. For each unit provide a light to indicate when each air conditioning system is in after hours mode.

Remote control panel shall be fully labelled. All labelling shall be engraved.

On the switchboard provide a green "RUN" and red "FAULT" light for each heater bank.

Meeting Room & Lunchroom Air Conditioning Systems (AC-1, AC-3, & AC-5)

Each air conditioning unit shall be complete with an automatic start-up and shutdown, 365-day time clock with holiday programming. Provide a building management system with a communication output and online access for monitoring and adjustment.

Air conditioning units to be provided with manufacturer's optional high level interface systems.

All controllers shall be grouped together in a switchboard panel within mech store 1.

All space thermostats shall be suitable for wall mounting. Temperature adjustment shall be from the BMS system only. A suitable dead band shall be incorporated before air conditioning unit operates in heating mode (reverse cycle).

The space sensor shall control the action of the supply air fan and compressor/s, so that the air conditioning unit can serve the zone at part load conditions by reduced air quantities and compressor capacity to limit operating within heating mode (reverse cycle).

Provide space on the switchboard, to allow addition of controls for four additional zones for each switchboard.

Provide a remote control panel where shown on the drawings to allow for out of hours operation. It shall be a switch, labelled 'off/auto'. In the 'auto' position the system shall operate via the time clock control. Where it is not shown, provide near main staff entry point.

For after hours operation of each unit, a labelled push button shall energise an adjustable 0 - 120 minute run timer to control the length of time that the plant operates in the after hours condition. Remote control panel shall be complete with green "ON" light and red "FAULT" light. For each unit provide a light to indicate when each air conditioning system is in after hours mode.

Remote control panel shall be fully labelled. All labelling shall be engraved.

CO₂ Monitoring (AC-1, AC-3, & AC-5)

Provide a CO₂ monitoring system to each air handling unit to automatically modulate the outside air according to the occupancy. Sensors shall be located within the served space adjacent the temperature sensor.

The monitoring system shall maintain the outside air flow rate within following limits:

Minimum:	50% of scheduled outside air flow
Maximum:	100% of scheduled outside air flow

The monitoring system shall be capable of being manually overridden to a position of full outside air flow as given in the schedule of minimum capacity.

Toilet Exhaust System

The toilet exhaust systems shall be controlled via an automatic start up and shutdown 365-day time clock with holiday programming installed in the Mechanical Services switchboard. Provide a green "RUN" and red "FAULT" light on the front of the switchboard.

25.0 ELECTRICAL

General

This Contractor shall supply and install all equipment and materials necessary for the complete and correct electrical operation of the mechanical services. Such equipment and materials shall include switchboards, control panels, motors, starters, contactors, fuses, relays, timers and wiring etc. as required.

All items of equipment shall be first grade with regard to design and manufacture and shall be completely satisfactory for operation, control, safety and maintenance under all conditions of service.

Uniformity of type and manufacture of switchgear, control gear, fittings and accessories shall be preserved throughout the whole of the installation.

Standards and Regulations

All electrical work shall be carried out strictly in accordance with the following:

- AS.3000 S.A.A. Wiring Rules and Amendments thereto;
- regulations of the Supply Authority.

The electrical contractor will provide power supplies as follows:

- three phase supply adjacent to each item of equipment;
- single phase supply adjacent to each item of equipment;

The mechanical contractor shall connect the supplies provided into the equipment, control panel or switchboard, and do all work on the load side of this point of connection as necessary for the complete installation and setting to work of equipment.

Prior to any work being carried out, this Contractor shall ascertain the size and type of supplies provided to his items of equipment, and ensure that the correct size power supplies are available. It shall be this Contractor's responsibility to upgrade any of the electrical feeds to his equipment if the required load check is not determined prior to the electrical cables being installed.

Package units and condensing units shall be provided with an isolating switch (lockable), installed adjacent to the unit but not on the unit, which isolates all parts of the system, including ancillary equipment.

VRF systems with separate power supplies to indoor and outdoor units shall have a warning label at the condensing unit, as per AS3000, and indoor units shall have independent isolating switches (lockable) installed adjacent each unit.

Air Conditioning Switchboards

Provide a switchboard in the location shown to control the mechanical services systems.

The design of the switchboard shall reflect the intent of the Electrical Safety Act with emphasis on all safety aspects of servicing and maintenance, and in particular, elimination of accidental contact with live terminals.

As a minimum:

- Main switches shall be fully shrouded.
- Bus bars shall not be exposed.
- Terminals on contactors and the like shall be fully recessed.
- The switchboard shall be lockable to prevent unauthorized access.

The switchboard shall:

- be floor or wall mounted and front connected;
- have main fault make isolating switch operable from the front of the switchboard (irrespective of whether the doors are open or shut);

- comply with the requirements of AS.3439.1 for Form 2 construction;
- be complete with main switch, starters, controls and protection and all other items as specified below, and as necessary for the control and operation of the installation;
- be manufactured or minimum 2.0mm zincanneal and be finished in baked enamel, white internally and electric orange externally;
- be provided with doors with neoprene gasket seal;
- be capable of withstanding the nominated fault current;
- be adequately labelled and provided with a typed circuit schedule in a perspex holder;
- be provided with suitably graded current limiting circuit breaker protection compatible with upstream protection to limit fault currents to the protected equipment.

Doors shall be of rigid construction, free from twists and warps. Hinges and locks or latches shall be brass, and attached by brass screws. Locks shall be spring type, provided with two keys, and unless otherwise agreed all locks on the one installation shall have identical keys.

Entry of cables, ducts and conduits shall be neatly made.

All mounting brackets and additional items shall be supplied and installed to suitably support the switchboard in the position in which it is to be erected.

All inside panel equipment shall be identified (and in the case of switches have their position marked) by means of engraved traffolyte labels showing black letters on white background. Block letters shall be used generally. All labels shall be securely fastened using chrome plated screws in such a manner that normal temperature variations do not cause distortion of the labels. Not less than two screws shall be used per label.

Switchboard in excess of 1.5m in length shall be provided with a switched fluorescent light and G.P.O.

Dust tight enclosures shall have ample volume to dissipate heat which may be generated in service, and door shall be provided with a neoprene seal fitted within a channel and closing against a suitable folded edge or ridge.

Separate earth and neutral bars shall be provided, with each terminal numbered, and all equipment shall be identified by screw-fixed traffolyte or engraved perspex labels. Adhesive tags will not be accepted.

General layout of control panels shall be submitted for approval.

Provide on the fascia of the switchboard, green "run" light and red "fault" light for each item of equipment controlled through the switchboard, including all heater banks plus a "Lamp Test" button.

Provide "auto-off-on" control switch for each air conditioning unit and ventilation fan.

Switches provided for control of the smoke fan shall be of an approved type and be lockable.

Starters

All motors of 15 kW or greater shall be voltage assisted start to give a starting transient of not more than four (4) times the running current. Those lesser than 15 kW shall be D.O.L. start.

In all conditions, such starters shall meet the local Supply Authority requirements having regard to starting current characteristics.

All starters shall incorporate thermal overload protection, and such other protective devices and auxiliary contacts as are necessary to the control and protection of the equipment.

Starters shall be selected for the duty and frequency of starts required, and arranged for operation in the order required for starting up and closing down plant. Operation of protective devices shall close down the plant in this sequence.

Wiring

Ample wiring space shall be provided within the control panel.

Internal power connections shall be by means of copper busbars of ample current rating and not less than 3 mm thick, and incoming and outgoing cables shall be attached to terminals by means of cable lugs. All live copper shall be insulated by P.V.C. tape, except at terminals which may need to be disconnected.

Note that no live metal shall be exposed except by removal of normally fixed panels.

Internal control wiring shall be P.V.C. insulated, using stranded conductors not smaller than 10 Amp rating. Conductors shall be neatly laid in cable trays. No loose wiring will be accepted.

Colour coding of the separate phases, neutral and earth, shall be provided and maintained throughout the installation. Where necessary further identification of wiring shall be provided, to the extent necessary, to permit a conductor to be located and traced.

Motors

Motors of 4 kW and above shall be provided with built in thermistors to protect all windings.

Method of Wiring

All wiring shall be carried out in P.V.C. conduit or where not acceptable as described below, in steel conduit.

P.V.C. Conduit

All wiring shall be carried out in heavy duty rigid P.V.C. conduit enclosing P.V.C. insulated conductors except that P.V.C. conduit shall not be installed in poured concrete or chased into walls not exposed within 1400 mm of the floor, nor where liable to severe mechanical damage, nor where exposed to direct sunlight.

P.V.C. conduit shall not be used where not permitted under the S.A.A. Wiring Rules, including hazardous locations and areas of high ambient temperature.

All bends shall be made by means of an internal bending spring, and where heat is applied, care shall be taken to avoid scorching of the surface. Cemented joints shall be made such that cement is still wet at time of making. Dry joints will not be accepted.

MIMS Conductors

MIMS conductors shall include but not be limited to, the earth sheathing return system.

Generally, single core MIMS cables 6 mm² or larger shall be installed using the ERS system.

Conductors shall be copper with copper sheathing, aluminium sheathing is unacceptable, and terminated with cold seat pots.

Conductors entering metal enclosures shall be terminated with suitable glands secured to electroplated copper or brass earthing plates.

Conductors shall be free of joints and tested immediately after termination and again after a 24 hour period, to indicate an infinity reading on a 1000 volt Megger.

Radox or similar may be used as an alternative, installation shall be in accordance with manufacturers instructions.

General

Wiring installed in plant rooms may be in surfaced mounted conduit, with surface boxes used for fittings, switches or outlets.

Elsewhere wiring shall be concealed throughout unless otherwise agreed, and all boxes for outlets, switches and the like shall be flush.

All wiring shall be segregated according to its working voltage run neatly and supported as per manufacturers detail. Runs greater than 12 circuits shall be on cable tray.

Conduits exposed to the weather shall be galvanised steel, and those embedded in concrete shall have not less than 20 mm of cover. Joints shall be painted with metallic paint.

Elbows and tees shall be avoided where practicable, and normal bends or sets used. Exposed outlet boxes shall be cast metal type, and flush boxes shall be cast or sheet metal.

Wires and Cables

All wires and cables shall be 660 volt rated insulation and be of new stock to the relevant S.A.A. Standard and shall arrive on the job with seals unbroken. Conductor material shall be copper, unless otherwise agreed and standard conductors only shall be used.

Cables shall be P.V.C. insulated unless otherwise specified and of ample size for the circuit loading and motor starting characteristics.

Active conductors shall be respectively red, white and blue and continuity of colour coding shall be preserved throughout the installation.

Terminations

All switchboard terminations shall be made to comply with AS 3000.

Terminals shall be numbered and individual terminal strips shall be prefixed with alphabetical character.

26.0 PAINTING AND CORROSION PROTECTION

The mechanical services installation does not require painting. Brass, bronze and copper components shall be left clean and polished.

Chrome plated, stainless steel and aluminium parts shall be left clean.

The standard paint finish on packaged equipment shall be maintained and left clean and polished at Practical Completion. Any damage to panels and paint finishes shall be repaired to a standard matching the original finish.

All items of equipment shall be identified by means of engraved traffolyte labels with black letters on a white background. Minimum 25mm height block letters only shall be used. All labels shall be securely fastened with minimum two (2) chrome plated screws.

27.0 TESTING AND COMMISSIONING

The whole of the contract works shall be in perfect working order on completion and shall, if necessary, be modified, repaired and adjusted to perform the required functions prior to practical completion.

Carry out all tests and analyses as are required both before and during the maintenance period to prove that the completed works comply with the contract requirements and shall be responsible for and supply all expenses in respect thereof.

Give reasonable notice in writing before any such tests are carried out.

Carry out all adjustments required and any specific adjustments and modifications as may be more particularly required by the contract specification, supply all necessary skilled labour for testing, operating and adjusting the systems.

Carry out commissioning tests to ascertain correct operation of the air conditioning and exhaust systems and supply all necessary labour for testing, operating and adjusting the systems.

The tests shall be scheduled so as to include tests prior to handover by this Contractor as well as additional testing within the warranty period so as to complete tests in ambient weather conditions ranging from hot humid "summer" conditions to "winter" conditions. This Contractor shall give at least forty eight (48) hours notice for each proposed test.

Results obtained during each test or part of test shall be logged and two copies of each log sheet shall be provided to the proprietor for record purposes.

The tests shall include but not necessarily be limited to the following:

- air quantity checks on 100% of air outlets on each ventilation or air conditioning system;
- temperatures using sling psychrometer to log dry bulb and wet bulb in representative locations for each separate air conditioning system;
- air temperatures shall be measured in air conditioning systems in the main supply air duct off the fan coil units;
- log all air flows, date, time, ambient dry bulb temperature, ambient wet bulb temperature;

In addition to the above, instruct the Proprietor or his representative fully in operations, adjustments and maintenance of all equipment and systems supplied under this contract.

Be fully responsible for the operation of all plant and equipment supplied and installed by him until the Proprietor acknowledges having received all necessary instructions in the operation of equipment and systems.

28.0 DEFECTS LIABILITY PERIOD

The defects liability period shall be fifty two (52) weeks for this contract and shall extend from the date of practical completion until the project defects liability period ends.

All work and materials shall be guaranteed as to quality and workmanship, and guaranteed against all defects for the duration of the defects liability period as defined above.

During this period promptly replace all defective fixtures and materials supplied and installed.

This Contractor shall also be responsible for all costs associated with damage to property and persons resulting from defective material, equipment and workmanship supplied under this contract until the end of the defects liability period.

29.0 SERVICE AND MAINTENANCE

Service and maintenance of equipment supplied and installed under this contract shall be carried out as follows during the defects liability period.

All maintenance, including maintenance by manufacturers and/or local suppliers shall be carried out in accordance with the appropriate maintenance manuals and as follows:

- a visit shall be made every month by the appropriate service engineer during the last week in the month;
- filters shall be checked and serviced at monthly intervals as recommended by the filter manufacturer;
- filters shall be replaced at not less than yearly intervals or as recommended by the filter manufacturer, and shall be replaced before the end of the Defects Liability Period;
- a register held by the Proprietor or his representative shall be signed by each service engineer;
- a monthly return, in triplicate, to the Proprietor shall be made by the first week in the following month stating maintenance work completed and action taken on defects by each service engineer;
- breakdown maintenance shall be attended to promptly on call from the proprietor and details entered on a monthly return;
- the proforma to be used and full details of maintenance procedure shall be to approval.

All minor lubricants and like consumable materials required for preventative maintenance shall be supplied by this Contractor.

30.0 SHOP DRAWINGS

Prepare shop drawings for review prior to any manufacture commencing. In preparation of the drawings, this Contractor shall co-ordinate his works with all building and other services.

The following drawings shall be submitted:

- ductwork layouts for air conditioning and exhaust systems;

- large scale details of conditioner unit, exhaust fan arrangement, duct and pipe support methods and any other information to indicate anti-vibration and acoustic treatments;
- details of associated builder work, e.g. roof, wall and floor penetrations, platforms and plant layouts;
- wiring diagrams and control diagrams.

At the completion of each portion of the installation, this Contractor shall ensure that drawings are corrected to an "as installed" condition.

31.0 OPERATION AND MAINTENANCE MANUALS

Operation and maintenance instructions shall be submitted, as drafts for examination, prior to final issue and shall be ready for issue prior to practical completion.

Three copies of the approved operation and maintenance manual shall be compiled by this Contractor for the Proprietor.

Manufacturer's advertising literature and/or advertising catalogues will not be acceptable for operation and maintenance instructions, but may be included for classification purposes.

Operating and maintenance instructions shall take generally the following form:

Professional Appearance

The manuals shall have a hard cover with name and title of the installation clearly engraved on it. It shall contain all material in A4 size to prevent cramping or binding. The manual shall be properly indexed and contain dividers, with tabs between each section. All data must be organised in a neat and concise manner. The sections shall consist of:

General Description of Systems and Equipment

Include brief overall description of system, design references and description of each individual system and equipment involved.

Maintenance of Systems and Equipment

Include maintenance duties in general, daily and all other periodic maintenance, lubrication chart and spare parts list. In addition, include a maintenance schedule as relevant to each component or group of components.

Commissioning Data

Equipment Schedules

Include schedules of equipment showing quantity, location, make, type, supplier, etc. a valve schedule and a schedule of all suppliers with addresses and telephone numbers.

Manufacturer's Literature

Include manufacturer's data on maintenance and operation of all equipment installed. Do not include irrelevant data.

Miscellaneous

Include any miscellaneous charts, graphs, description, data, etc., needed for complete maintenance and operating instructions of all systems and equipment installed.

Each manual shall have a set of approved "as installed" drawings, neatly bound into the manual with cover sheet and index.

Provide an electronic copy of the Operating and Maintenance Manuals and "As Installed" drawings on CD.

32.0 SCHEDULE OF EQUIPMENT MINIMUM CAPACITIES

Refer to the drawings for Schedules of Capacities for all equipment.